

Canon R7 EXR Converter - FVFX

User manual · version 2.2.0

What it does

The program converts video clips recorded on a Canon EOS R7 in Canon Log 3 into linear OpenEXR image sequences. Each clip becomes a folder with one EXR file per frame. The files can be loaded directly into Nuke.

For every frame the program:

Step	What happens
1. Decode	Reads the 10-bit 4:2:2 HEVC video from the camera file with ffmpeg.
2. Linearise	Converts the Canon Log 3 curve to linear light (18% grey = 0.18).
3. Colour	Converts from the camera's colour space (Cinema Gamut, read from the clip) to ACEScg.
4. Write	Saves a 16-bit half-float EXR with ZIP compression, frame rate, timecode, camera and lens information.

Getting started

Runs on 64-bit Windows 10 or 11. Nothing else needs to be installed.

Installer: run Canon R7 EXR Converter v2.2.0 Setup.exe. No admin rights are needed. Start the program from the Start menu.

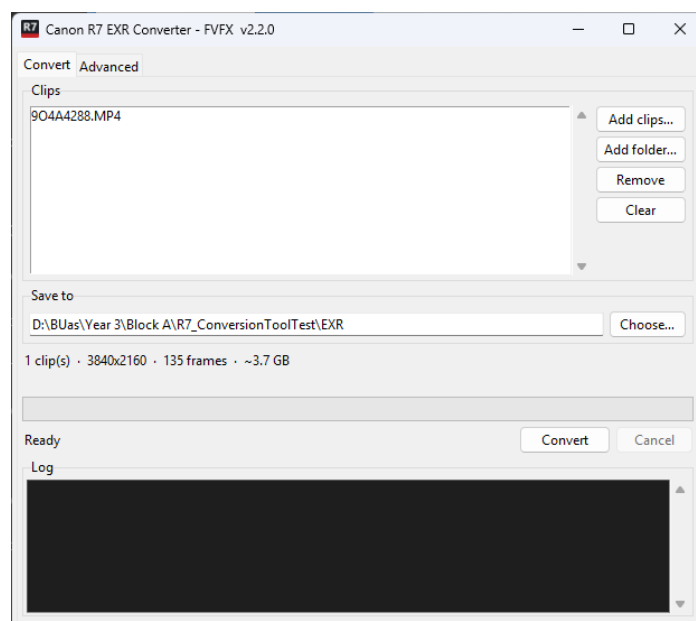
Zip: alternatively, extract the zip to a normal folder, for example Documents (keep all the files together), and run Canon R7 EXR Converter v2.2.0.exe.

The first time, Windows may show “**Windows protected your PC**”. Click **More info** and then **Run anyway**.

Converting clips

1. Click **Add clips...** to pick files, or **Add folder...** to add every video in a folder.
2. Check the **Save to** folder. It defaults to an **EXR** folder next to the first clip. Click **Choose...** to change it.
3. Click **Convert**. Progress is shown in the bar and the log. A message appears when all clips are done.

The line under the folder shows the number of clips, resolution, total frames and the estimated size on disk before you start.



Output

Each clip is written to its own folder, named after the clip:

```
EXR\904A4288\904A4288.0001.exr
EXR\904A4288\904A4288.0002.exr
...
```

Opening in Nuke

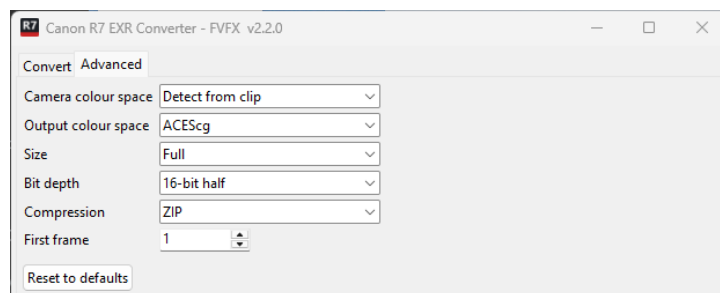
The frames are scene linear ACEScg. Set Nuke up for ACES once per script:

1. Open **Edit** → **Project Settings** (or press **S** in the Node Graph) and click the **Color** tab.
2. Set **color management** to **OCIO**.
3. In **OCIO config**, choose an ACES 1.3 config. The working space becomes scene_linear (ACEScg).
4. Set the Read node's colourspace to ACEScg.

Do not use Nuke's legacy colour management (nuke-default) with this footage, and do not convert to **Linear Rec.709**: much of the camera's wide gamut falls outside Rec.709. **Linear, camera primaries** is only for pipelines that want Cinema Gamut untouched (read as Linear CinemaGamut D55). Details: Foundry's OCIO Color Management page in the Nuke online help.

Advanced settings

The Advanced tab holds settings that normally do not need changing. **Reset to defaults** restores all of them.



Setting	Default	Options
Camera colour space	Detect from clip	Detect from clip, Canon Cinema Gamut, BT.709 / sRGB, BT.2020
Output colour space	ACEScg	ACEScg; Linear, camera primaries; Linear Rec.709
Size	Full	Full, 1920 wide, 1280 wide (aspect ratio is kept)
Bit depth	16-bit half	16-bit half, 32-bit float
Compression	ZIP	ZIP, PIZ, ZIPS, DWAA (lossy), None
First frame	1	Any whole number

File size

Measured on R7 4K footage, per frame and for a 5.4 second (135 frame) clip:

Setting	Per frame	5.4 s clip	Lossless
16-bit, ZIP (default)	29.6 MB	3.7 GB	Yes
16-bit, PIZ	27.4 MB	3.4 GB	Yes
16-bit, ZIPS	30.4 MB	3.8 GB	Yes
16-bit, DWAA	7.7 MB	1.0 GB	No
16-bit, None	49.8 MB	6.3 GB	Yes
32-bit, ZIP	81 MB	11 GB	Yes

16-bit, ZIP, 1920 wide	7.4 MB	0.9 GB	Yes
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Messages

Message	Meaning
ffmpeg was not found	Files are missing from the program folder. Extract the whole zip again.
exiftool not found	Files are missing from the program folder. Canon Cinema Gamut is used until the zip is extracted again.
Colour space not found in clip	The clip has no Canon colour space tag. Canon Cinema Gamut is used. Set Camera colour space manually if this is wrong.
Clip is ..., not Canon Log 3	The clip was recorded with a different picture profile. The output colours will be incorrect.
Could not read ...	The file is not a readable video file.

Technical reference

Item	Value
Input	HEVC (H.265) Rext, 10-bit 4:2:2, Canon Log 3, as recorded by the EOS R7
Log decode	Canon Log 3 published curve, legal-range code mapping $((code - 64) / 876)$
Colour conversion	3×3 matrix from the camera gamut to ACEScg by default, or to Rec.709 (Bradford adaptation where white points differ). Optionally none (camera primaries kept)
Resizing	Lanczos, applied in linear light
Libraries	FFmpeg 8.1 (decode, LGPL), ExifTool (Canon metadata), numpy, OpenImageIO (EXR writing), tkinter (interface). Built with PyInstaller.